EE2004: Digital Signal Processing

Tutorial 2

(February 5, 2026)

1. The discrete-time sinusoids given below are either in the form $\cos(\omega n + \theta)$ or $\sin(\omega n + \theta)$, where ω can be any finite value. Express each signal in its corresponding alternate form, i.e., either $\cos(\omega_1 n + \theta_1)$ or $\sin(\omega_1 n + \theta_1)$, where $\omega_1 \in [0, \pi]$. (a) $\sin(1.6\pi n + \theta)$, (b) $\cos(2.3\pi n + \theta)$, (c) $\cos(34.6991 n + \theta)$, and (d) $\sin(-2.3\pi n + \theta)$.

2. A real-valued sinusoid has frequency $\omega = 2\pi,\ 3\pi,\ 5\pi,\ 3.2\pi,\ 22.1327,\ \pi + 2$. To produce an identical signal in each case, what should the corresponding frequency be if it is restricted to the range $[0, \pi]$? In which cases does the phase change sign?

3. What values of θ cause the DT sinusoid $\cos(\omega_0 n + \theta)$ to be a simple shifted version of $\cos(\omega_0 n)$?

4. Find the simplified expression for $x[n]$ where $x[n] = \cos(0.6\pi n + \frac{\pi}{6}) + \sqrt{3}\cos(1.4\pi n + \frac{\pi}{3}) - 2\cos(0.6\pi n - \frac{\pi}{6})$.

5. Express the following exponentials in the form $e^{j(\omega_0 n + \theta)}$ where $\omega_0 \in [-\pi, \pi)$: (a) $e^{j(8.2\pi n + \theta)}$, (b) $e^{j4\pi n}$, (c) $e^{-j1.95 n}$, and (d) $e^{-j10.7\pi n}$. Repeat the problem if we require $\omega_0 \in [0, 2\pi)$.

6. Consider $x[n] = \cos(\omega_0 n)$ for various values of ω_0 in the set $\{0, 0.1\pi, 0.2\pi, 0.8\pi, 0.9\pi, \pi, 1.1\pi, 1.2\pi, \sqrt{2}\}$. What is the period of $x[\cdot]$ in each case?

7. Determine the period of $x[n] = \sin(0.08\pi n)$. List two other *independent* sinusoids that have the same period as $x[n]$.

8. Consider the harmonically related periodic signals $x_k[n] = \cos(2\pi k n / N)$, where $0 \leq k \leq N - 1$. Since $x_k[n]$ is real-valued, the highest value of k can be limited to the integer that is less than or equal to $N/2$. What is the period of $x_k[n]$ for each k when $N = 11$ and $N = 12$? Is there any difference in behaviour when compared with its continuous-time counterpart?

9. A system may or may not be (a) memoryless, (b) time-invariant, (c) linear, (d) causal, (e) stable. Determine which of these properties hold and which do not hold for each of the systems whose input-output relationships are given below. Justify your answers. In each example $x[n]$ the system input and $y[n]$ denotes system output. (i) $y[n] = \text{Ev}\{x[n-1]\}$ (where $\text{Ev}\{\cdot\}$ represents the even part),
(ii) $y[n] = \begin{cases} x[n] & n \geq 0 \\ x[n+1] & n \leq -1 \end{cases}$, (iii) $y[n] = x[2n]$.

10. Determine whether or not each of the following systems is linear and time-invariant; justify your answer. (a) $y[n] = e^{x[n]}$, (b) $y[n] = x[-n]$, and (c) $y[n] = x[n] + 3u[n+1]$.

11. For each system give below, determine which of the following attributes hold: linearity, causality, time-invariance. Wherever necessary assume zero initial conditions. If a property is true, prove it; otherwise, give a counterexample.

(a) $y[n] = y[n-3] + 10x[n]$
(b) $y[n] = \min_{0 \le m \le n} \{x[m]\}$
(c) $y[n] = 2^{|n|}\, x[n]$
(d) $y[n] = \Re\mathrm{e}\{x[n]\}$
(e) $y[n] = \cos(\pi n) + x[n-4]$
(f) $y[n] = \begin{cases} 1 & \text{if } x[n] \ge 0 \\ -1 & \text{otherwise} \end{cases}$
(g) $y[n] = x^*[n]$,
(h) $y[n] = -2|x[n]| + \sum_{m=n-2}^{n+2} x[m]$

12. Find the simplified expression for $y = x * h$, where $x[n] = a^{-n}u[n]$, $h[n] = -b^n u[-n-1]$, with $|a| \neq |b|$ and $|a|, |b| > 1$.

13. Verify the following properties of convolution:

Signal Type	$x[n] = 0$ for	$h[n] = 0$ for	$y[n] = 0$ for	Duration
Right sided	$n < N_1$	$n < N_2$	$n < N_1 + N_2$	
Left sided	$n > N_1$	$n > N_2$	$n > N_1 + N_2$	
Finite Duration	$n < N_1$ $n \ge N_3$ $N_x = N_3 - N_1$	$n < N_2$ $n \ge N_4$ $N_h = N_4 - N_2$	$n < N_1 + N_2$ $n > N_3 + N_4 - 1$	$N_y = N_x + N_h - 1$

14. Simplify, if possible, the following convolutions: (a) $u[n] * u[n]$, and (b) $u[n] * u[-n]$.

15. The *cross-correlation* of two real-valued signals $x[n]$ and $w[n]$ is defined as

$$\phi_{xw}[k] = \sum_{n=-\infty}^{\infty} x[n+k]\, w[n]$$

The special case when $x[n] = w[n]$ is called the *autocorrelation* function.
(a) Express $\phi_{xw}[k]$ as a convolution.
(b) Find the relationship between $\phi_{xw}[k]$ and $\phi_{wx}[k]$ and also between $\phi_{xx}[k]$ and $\phi_{xx}[-k]$.
(c) Let $x[n] = u[n] - u[n-N+1]$. Plot (a) $y[n] = x[n] * x[n]$, and (b) $\phi_{xx}[k]$.

16. (a) Working in the time-domain and using a step-by-step procedure, assuming $x[n] = \delta[n]$, find the output the system described by the following difference equation: $y[n] - 0.8\, y[n-1] = x[n]$. Assume $y[-1] = 0$.
(b) Repeat when the input is $0.5\, u[n]$; as before, assume $y[n] = 0$ for $n < 0$. Compare this answer with the result of $0.5\, u[n] * 0.8^n\, u[n]$? Can you relate this to the system output obtained in part (a) ?

17. $y[n] = x[n] * h[n]$. Let $h[n] = 0$ outside $0 \le n \le N-1$, which reduces the convolution summation to a finite one. For each n, the output $y[n]$ can be thought of as an *inner product* of two N-dimensional vectors, i.e., $\mathbf{a}^T\mathbf{b}$. Let $\mathbf{y}$ represent in vector form the output from 0 to $N-1$. The convolution can be expressed as $\mathbf{y} = \mathbf{X}\mathbf{h}$. What are the entries of $\mathbf{X}$ and $\mathbf{h}$? What can you say about the diagonal entries of $\mathbf{X}$?

18. Suppose $y = x * h$. $\{x\} = \{2, 2, 3, 3, 4\}$ and $\{y\} = \{2, 4, 9, 10, 13, 10, 8\}$. Find $\{h\}$.

19. If you multiply two polynomials, the coefficients of the result can be obtained by considering each polynomial's coefficients as a sequence and convolving the two. Consider $\{1, 3, 1\} * \{1, 2, 2\}$. Use this result to compute 131×122.

20. Consider an LTI system with $h[n] = 2^{-n}\, u[n]$. Find the output $y[n]$ if the input $x[n] = u[n] - u[n-10]$.

21. **Computer assignment** Consider the convolution of $\{1, 2, 3, 4\}$ and $\{3, 2, 1\}$. Determine the 6×4 matrix $\mathbf{H}$ such that $\mathbf{y} = \mathbf{H}\mathbf{x}$. Since convolution is commutative, obtain $\mathbf{y}$ by using the equation $\mathbf{X}\mathbf{h}$. What are the entries of $\mathbf{X}$? A convolution matrix can be quickly constructed using the `toeplitz` command; see also `convmtx`.

22. **Computer assignment** Let $\mathbf{y} = \mathbf{X}\mathbf{h}$, where the matrices are as defined in the previous problem. Consider the sequences given in Problem 18. Let $\mathbf{h} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$. Does this result agree with what you obtained earlier? Look up material on the least-squares solution of $\mathbf{A}\mathbf{x} = \mathbf{y}$. Can you obtain $\mathbf{h}$ by solving $\mathbf{y} = \mathbf{X}\mathbf{h}$ recursively?

23. **Computer Experiment** To generate a sequence with frequency components 0.1 and 0.2, first define `n=(0:99)';` to get a column vector of length 100. The sequence `x` can then be generated using the command `x = cos(2*pi*0.1*n+pi/3) + sin(2*pi*0.2*n + pi/8);`. The higher frequency component can be eliminated by convolving `x` with `h`, where `h = 0.2*ones(5,1);`. Plot `y` where `y = conv(x,h)`. Verify that the second component is practically absent in `y`. Give a theoretical explanation as to why the convolution with an all ones sequence of length 5 has eliminated the sinusoid with frequency 0.2.

24. **Computer Experiment** Form the sequence $h[n] = a^n\, u[n]$ for $a = 0.9$. Let $x[n] = \sin(2\pi n/16)$ for $n = 0, 1, ..., 999$. Compute the convolution of these two sequences and plot the first hundred samples. The built-in commands for convolution are `convol` (Scilab) and `conv` (Matlab).

    Calculate the theoretical convolution output when $h[n] = (0.9)^n\, u[n]$ is convolved with $x[n] = \sin(2\pi n/16)\ \forall n$. The result can be decomposed into two terms: $\alpha\, 0.9^n\, u[n] + \beta \sin(2\pi n/16 + \gamma)$. Calculate the constants α, β, and γ and compare the theoretical results with that obtained from Matlab/Scilab.

25. **Computer assignment** Consider the convolution of $x[n]$ and $y[n]$, where $y[n] = x[-n]$. The result is called the *autocorrelation* of $x[n]$. Let `x[n] = rand(100,1)`. Find the autocorrelation of $x[n]$. Normalize the peak value to unity. At what time index does it occur? Replace `rand` by `randn` and repeat. Do you see any qualitative change in the result? Study the function `xcorr`. If the samples of a sequence are uncorrelated, the resulting autocorrelation sequence will be impulsive (or approximately so). For example, consider N successive digits of π. Replace an even digit by 1 and and odd digit by 0. Find the autocorrelation of the resulting sequence. The first ten thousand digits of π can be found at `http://www.joyofpi.com/pi.html`. Repeat the experiment by replacing the even digits by 1 and the odd-digits by -1. What is the main difference that you observe in this case? $\diamond$